Laboratory Assignment 5: Optimization Implementations of Logic Functions

ECE 0201: Digital Circuits and Systems

(45 Points)

Name

张鹤扬 Stefan

Submission Checklist:

- ☐ Write within boxes, do not move boxes
- ☐ Write your full name in the box above
- ☐ Save this file as a PDF before uploading, keep the number of pages (6) unchanged
- ☐ Note "TO BE CONTINUED" in the answer box if you used the extra page (6)

Part II: Getting Started

[(A) Define your abstraction and create a truth table] (4 points)

1. Definition:

condition	Error	Paper	Scissor	Rock
X_1X_0	11	00	01	10

Note: X can be A or B.

2. Abstraction:

Switch open is 0, close is 1;

LED ON is 1, OFF is 0

3. Truth table:

A_1A_0	B_1B_0	W_A	W_B	E
00	00	1	1	0
00	01	0	1	0
00	11	-	-	1
00	10	1	0	0
01	00	1	0	0
01	01	1	1	0
01	11	-	-	1
01	10	0	1	0
11	00	-	-	1
11	01	-	-	1
11	11	-	-	1
11	10	-	-	1
10	00	0	1	0
10	01	1	0	0
10	11	-	-	1
10	10	1	1	0

* **-** are **don't care the value**

[(B) Explain how you have chosen to indicate a tie, and why you made this design choice (3 points)]

Indication: A tie is represented by both WA and WB being asserted (lighted) simultaneously.

Reason: Treating the tie condition as WA = WB = 1 introduces additional minterms into the K-maps for WA and WB. This increases grouping opportunities, thereby yielding simpler (fewer literals or gates) sum-of-products expressions.

[(C) Write and simplify a logic function for each output (show K-maps if you choose to use them)] (4 points)

1. Error: From the truth table, E = 1 whenever A equals 11_2 (i.e., A1A0 = 11) or B equals 11_2 (i.e., B1B0 = 11).

$$E = A_1A_0 + B_1B_0$$

2. W_A: WA must be 1 whenever A wins or when the scores are tied. Additionally, whenever the error condition occurs (E LED on), the states of WA and WB are irrelevant and may be treated as don't-care ("X"). This allows us to fill X's in the corresponding cells of the WA K-map.

A_1A_0 B_1B_0	00	01	11	10
00	1	0	X	1
01	1	1	X	0
11	X	X	X	X
10	0	1	X	1

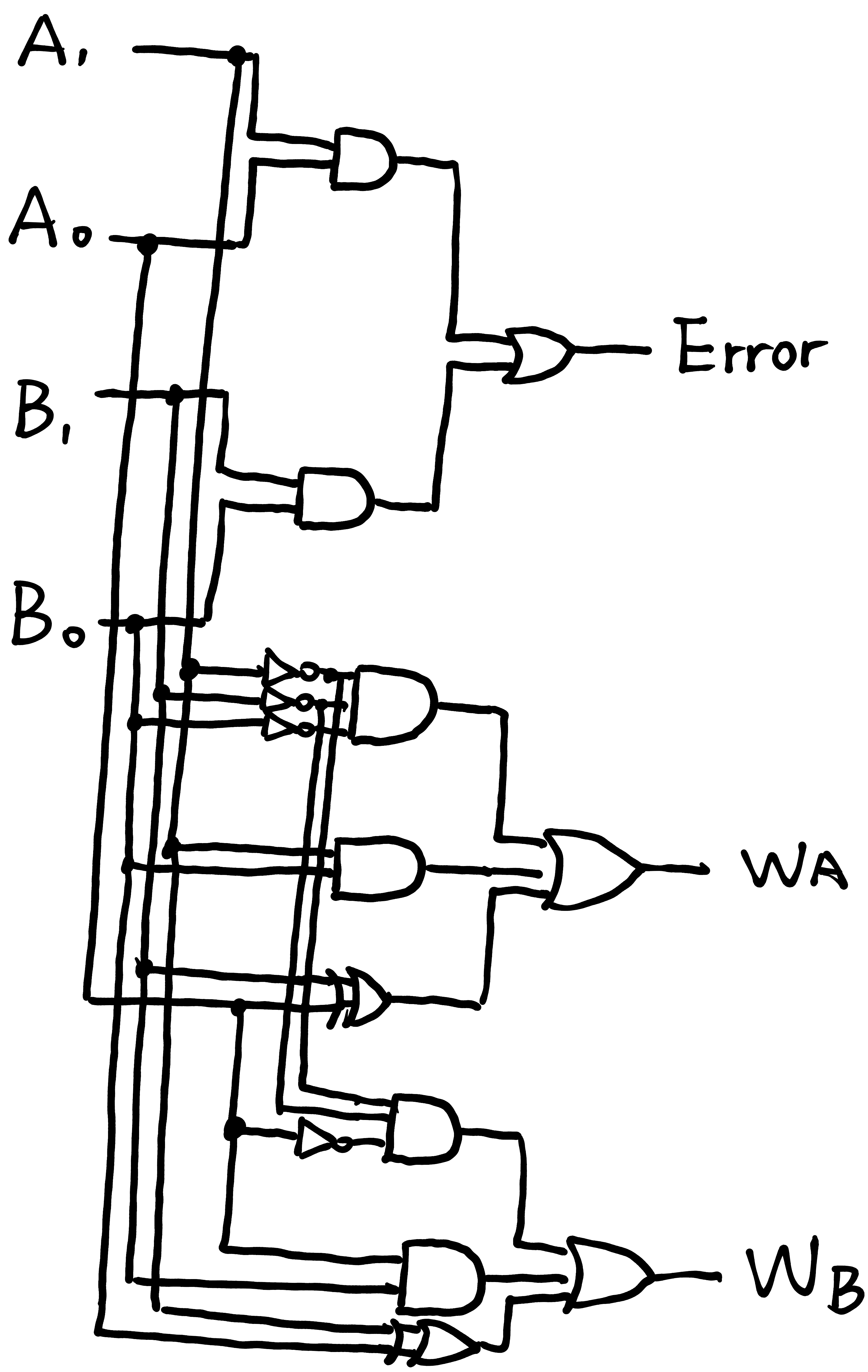
$$W_A = A_1'B_1'B_0' + A_1B_0 + A_0 \oplus B_1$$

3. W_B: WB must be 1 whenever B wins or when the scores are tied. Similarly, under the error condition (E LED on), the states of WA and WB become irrelevant and are marked as don't-care ("X") in the WB K-map.

A_1A_0 B_1B_0	00	01	11	10
00	1	1	X	0
01	0	1	X	1
11	X	X	X	X
10	1	0	X	1

$$W_B = A_1'A_0'B_1' + A_0B_1 + A_1 \oplus B_0$$

[(D) Draw a gate-level schematic that implements your logic functions and insert a picture of it here] (4 points)



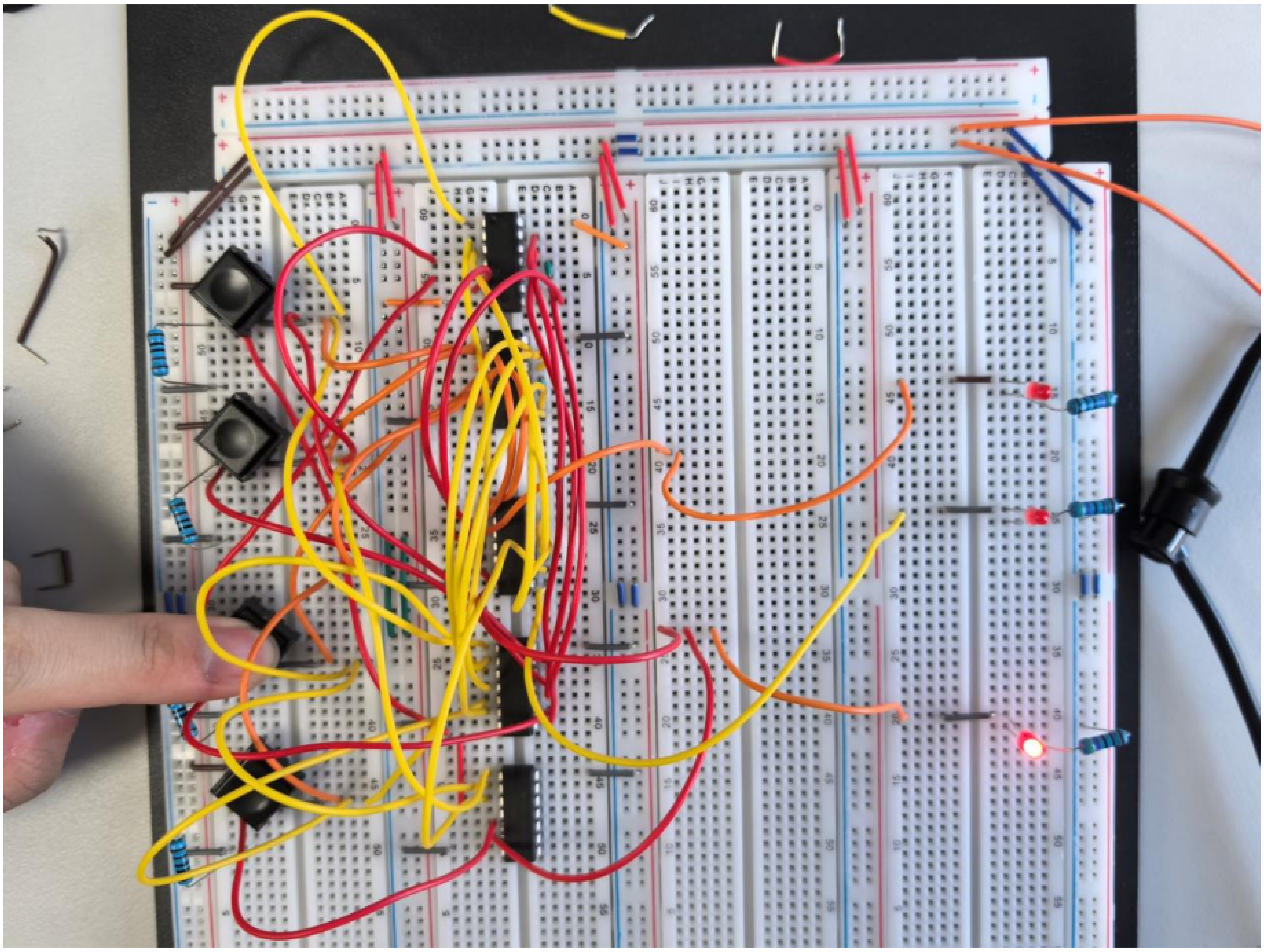
(E) (5 points)

How many gates would the **canonical design** require? 9

How many gates does your design require? 17

How many IC Chips did you use? 5

[(F) Build the circuit and insert a picture of it here] (5 points)



[(G) Demonstrate the functionality of your circuit to a TA] (20 points)

[Leave the page unchanged if you are not going to use. Otherwise, specify the problem number you are working on here. Don't forget to notify "TO BE CONTINUED" in the original problem box]